

Generalization of a Limit Involving the Central Binomial Coefficient

Johar M. Ashfaque

August 6, 2026

The Original Limit

The problem presented in `dailyintegral-limit-limit-1x1-2026-03-17.jpg` asks to evaluate:

$$L = \lim_{x \rightarrow \infty} x \left(\frac{\sqrt{\pi x} \binom{2x}{x}}{4^x} - 1 \right) \quad (1)$$

To evaluate and subsequently generalize this limit, we first express the central binomial coefficient in terms of the Gamma function. Using Legendre's duplication formula,

$$\Gamma(2x) = \frac{4^x}{2\sqrt{\pi}} \Gamma(x) \Gamma\left(x + \frac{1}{2}\right), \quad (2)$$

we can rewrite $\binom{2x}{x}$ as:

$$\binom{2x}{x} = \frac{\Gamma(2x+1)}{\Gamma(x+1)^2} = \frac{2x\Gamma(2x)}{x^2\Gamma(x)^2} = \frac{4^x}{\sqrt{\pi}} \frac{\Gamma\left(x + \frac{1}{2}\right)}{\Gamma(x+1)}. \quad (3)$$

Substituting this relation back into the original bracketed expression yields:

$$\frac{\sqrt{\pi x} \binom{2x}{x}}{4^x} = \sqrt{x} \frac{\Gamma\left(x + \frac{1}{2}\right)}{\Gamma(x+1)}. \quad (4)$$

The original limit thus takes the equivalent continuous form:

$$L = \lim_{x \rightarrow \infty} x \left(x^{1/2} \frac{\Gamma\left(x + \frac{1}{2}\right)}{\Gamma(x+1)} - 1 \right). \quad (5)$$

A Rigorous Generalization

The rewritten limit is a specific instance of a broader asymptotic property of the Gamma function. We can define the generalized limit as:

$$\mathcal{L}(a, b) = \lim_{x \rightarrow \infty} x \left(x^{b-a} \frac{\Gamma(x+a)}{\Gamma(x+b)} - 1 \right) \quad (6)$$

for real parameters a and b .

To evaluate $\mathcal{L}(a, b)$, we rely on the asymptotic expansion of the ratio of two Gamma functions (often derived from Stirling's series or associated with Tricomi and Erdélyi). Expanding the natural logarithm of the Gamma function gives:

$$\ln \Gamma(x + a) = \left(x + a - \frac{1}{2}\right) \ln x - x + \frac{a^2 - a}{2x} + \mathcal{O}\left(\frac{1}{x^2}\right) \quad \text{as } x \rightarrow \infty. \quad (7)$$

Taking the difference $\ln \Gamma(x + a) - \ln \Gamma(x + b)$ yields:

$$\ln \left(\frac{\Gamma(x + a)}{\Gamma(x + b)} \right) = (a - b) \ln x + \frac{(a - b)(a + b - 1)}{2x} + \mathcal{O}\left(\frac{1}{x^2}\right). \quad (8)$$

Exponentiating this result provides the asymptotic series for the ratio itself:

$$\frac{\Gamma(x + a)}{\Gamma(x + b)} = x^{a-b} \exp \left(\frac{(a - b)(a + b - 1)}{2x} + \mathcal{O}\left(\frac{1}{x^2}\right) \right). \quad (9)$$

Using the Maclaurin series expansion for the exponential function, $\exp(z) = 1 + z + \mathcal{O}(z^2)$, we obtain:

$$\frac{\Gamma(x + a)}{\Gamma(x + b)} = x^{a-b} \left(1 + \frac{(a - b)(a + b - 1)}{2x} + \mathcal{O}\left(\frac{1}{x^2}\right) \right). \quad (10)$$

Multiplying by x^{b-a} and rearranging the terms gives:

$$x^{b-a} \frac{\Gamma(x + a)}{\Gamma(x + b)} - 1 = \frac{(a - b)(a + b - 1)}{2x} + \mathcal{O}\left(\frac{1}{x^2}\right). \quad (11)$$

Finally, multiplying by x and taking the limit as $x \rightarrow \infty$ causes the higher-order terms to vanish, producing the generalized result:

$$\mathcal{L}(a, b) = \frac{(a - b)(a + b - 1)}{2}. \quad (12)$$

Evaluating the Specific Limit

Returning to the original problem, we can clearly identify the parameters as $a = \frac{1}{2}$ and $b = 1$. Substituting these values into our generalized formula yields:

$$L = \mathcal{L}\left(\frac{1}{2}, 1\right) = \frac{\left(\frac{1}{2} - 1\right)\left(\frac{1}{2} + 1 - 1\right)}{2} = \frac{\left(-\frac{1}{2}\right)\left(\frac{1}{2}\right)}{2} = -\frac{1}{8}. \quad (13)$$

Thus, the limit rigorously converges to $-\frac{1}{8}$.